

Implied Odds

Does Polymarket price Bitcoin volatility like Deribit? A 75-day study of daily BTC price markets.

Ivan Alonso H.

Independent crypto market research

Founder, BCSentinel AI · MSc Data Science & AI (in progress) · Barcelona

github.com/ivanalonso-dev · linkedin.com/in/ivanalonsobcsentinel · x.com/IvanAlonsoBCSen

Executive summary

Polymarket runs a daily market on where Bitcoin will close, split into \$2,000 price brackets that each pay \$1 if BTC lands inside. A set of brackets is in effect a strip of digital options, so its prices imply a volatility forecast. This study extracts that forecast and tests it against the professional benchmark, Deribit's DVOL index, over 75 days (14 July – 5 October 2026).

- **A day ahead, the crowd forecasts better than the options desk.** At 24 h and 12 h before the close, Polymarket's bracket probabilities are significantly more accurate (Brier score, paired $t > 2$) than probabilities built from DVOL, from DVOL adjusted for time of day, and from recent realized volatility.
- **Polymarket's volatility has a term structure; DVOL is flat.** About 31% annualized a day out and about 47% in the final hour, consistent with BTC moving most in the US session. DVOL, a 30-day measure, stays near 37%.
- **In the last hour, it's a tie.** All models score the same.
- **No measurable lag.** Minute by minute, Polymarket tracks the BTC price with no lag of one minute or more.

1. Background

Deribit, options and DVOL

Deribit is the world's largest crypto options exchange, handling the vast majority of Bitcoin and Ethereum options volume, mostly from professional traders. Coinbase completed its acquisition in 2025. An option gives the right, but not the obligation, to buy or sell BTC at a set price on a future date; its price depends mainly on how much the market expects BTC to move.

DVOL is Deribit's index derived from its option prices: it measures expected annualized volatility over the next 30 days, the crypto equivalent of the VIX. A DVOL of 45 implies an expected move of about 2.4% per day.

Why compare Polymarket with Deribit

Polymarket's daily brackets are options too: each pays \$1 if BTC ends inside a range, so their prices imply an expected volatility. The question is simple: **does Polymarket's crowd expect the same volatility as Deribit's professionals?** If not, one of them is wrong, and the data shows which.

2. Data and method

1. **Data.** Daily BTC markets from Polymarket's public APIs (bracket bounds, settlement and minute-level prices), Deribit DVOL (hourly) and Binance BTCUSDT candles. The settlement rule is the close of the Binance 1-minute candle at 12:00 ET; our reconstruction matches the official winner on 100% of days.

2. **Snapshots** at 24, 12, 6, 3 and 1 h before the close. A snapshot is used only if every bracket price is less than 3 minutes old.
3. **Implied volatility.** Prices are normalized to sum to 1 (removing the 1–2% overround) and a driftless lognormal centered on the Binance spot is fitted. The volatility that best reproduces the prices is Polymarket's implied volatility.
4. **Benchmarks**, all using only data available at the snapshot: raw DVOL; DVOL reweighted by the historical variance of each hour of the day; and 7-day realized volatility with the same adjustment.
5. **Scoring.** Brier score and log-loss over all brackets, compared day by day with a paired t-test.
6. **Lag test.** Polymarket prices at minute t against fair bracket probabilities from the BTC price at $t - k$, $k = 0 \dots 10$ minutes, with timestamps aligned so that any bias favors Polymarket.

3. Results

How much volatility each side expects

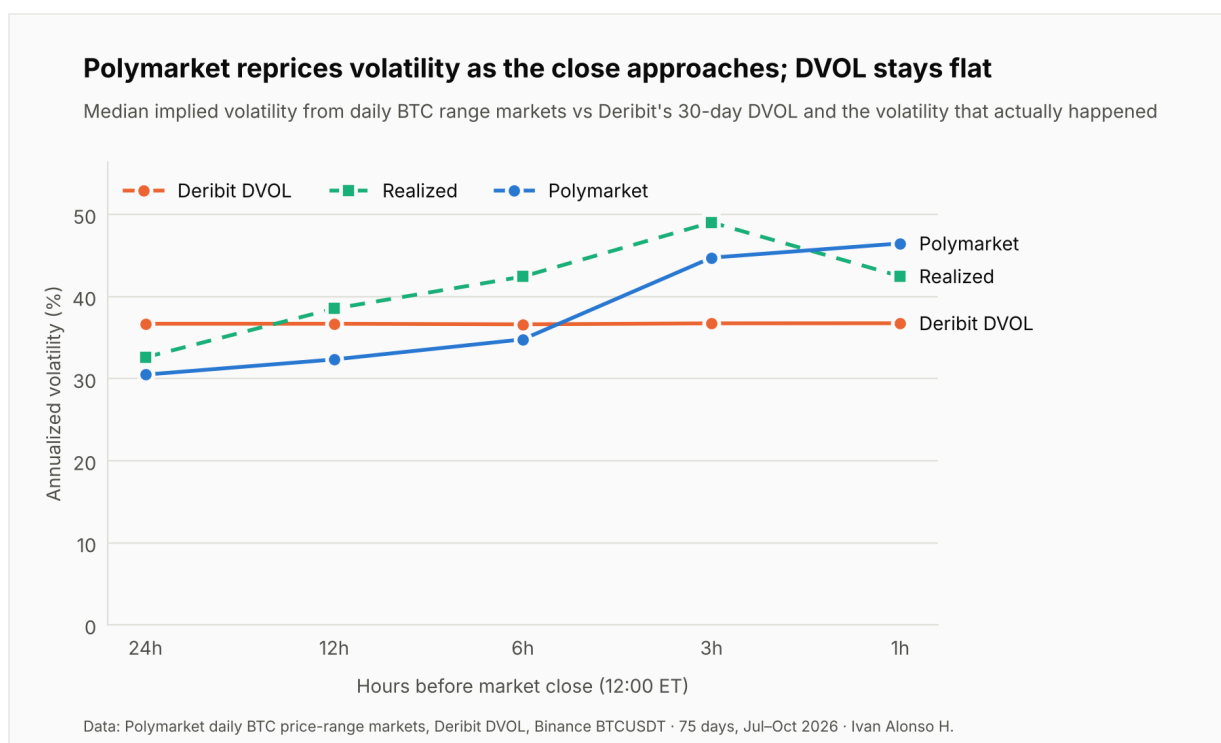


Figure 1. Polymarket's median implied volatility vs DVOL and the volatility that actually happened, by hours to close.

At 24 h, Polymarket expected 31% and realized volatility was 33%; DVOL stood at 37%. As the close approaches, Polymarket raises its estimate, tracking BTC's intraday pattern. DVOL can't, because it measures a 30-day horizon.

Who is more accurate

Hours to close	Polymarket	DVOL	t	DVOL, time adj.	t	Realized 7d	t
24 h	0.499	0.537	2.76	0.537	2.76	0.524	2.30
12 h	0.414	0.448	3.09	0.460	3.05	0.440	2.05
6 h	0.371	0.388	1.45	0.400	1.83	0.398	2.33
3 h	0.338	0.360	1.87	0.355	1.19	0.358	1.81
1 h	0.220	0.217	-0.30	0.221	0.06	0.215	-0.36

Mean Brier score over all brackets (lower is better), 75 days. t = paired t-statistic of the difference vs Polymarket; t > 2 means Polymarket is significantly better.

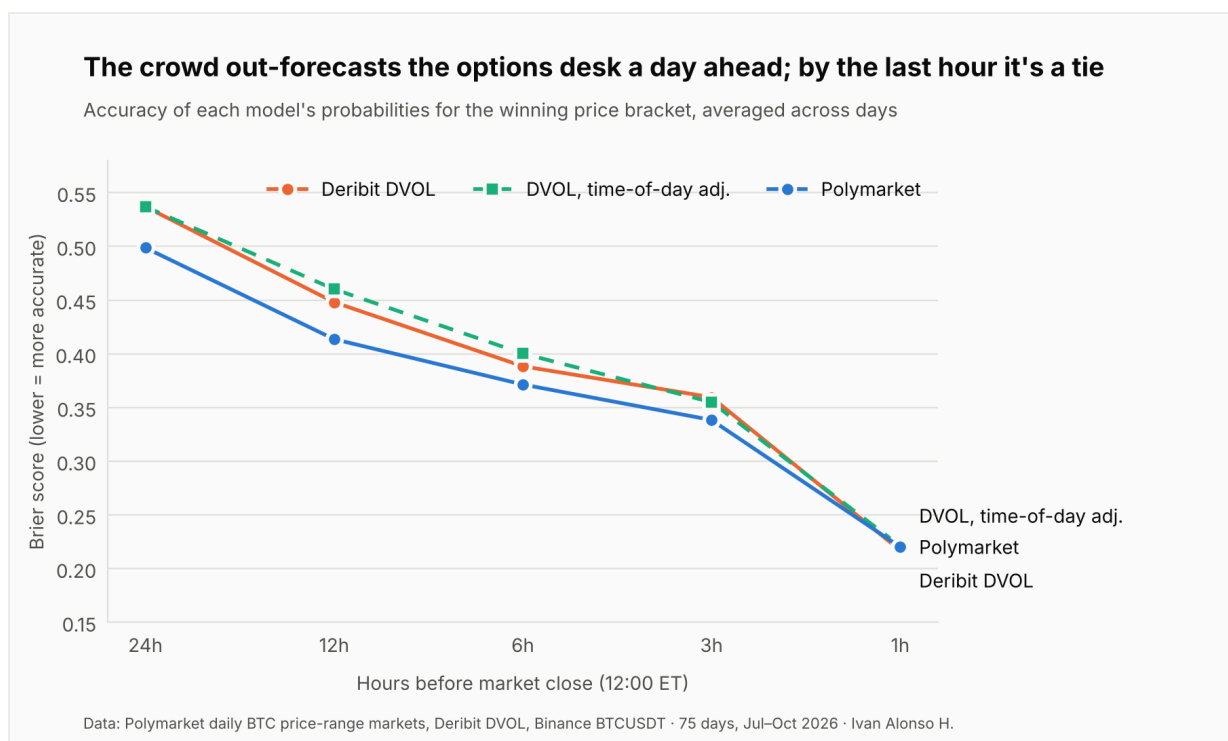


Figure 2. Accuracy of each model by hours to close.

At 24 h and 12 h, Polymarket beats all three benchmarks with statistical significance. At 6 h and 3 h it is still better, though not significantly against every benchmark. In the final hour all four models tie.

Does Polymarket lag the price?

Using minute-level data for the last hour of 79 days, the smallest error between Polymarket and the fair probability occurs at zero lag, and it grows steadily as more lag is assumed. On 62 of 79 days the best fit is "no lag". Lags of seconds, where trading bots operate, can't be measured with this data.

4. Limitations

- **Sample:** 75 days in a single volatility regime (DVOL between 33 and 45). Results at 6 h and 3 h point the same way but are not significant against every benchmark.

- **Mid prices, not executable prices.** Historical prices are midpoints; the order book is not archived, so trading profitability can't be tested.
- **Model:** a lognormal centered on spot. Any directional view held by the crowd would be partly absorbed into the fitted volatility.
- **DVOL horizon:** DVOL is a 30-day index. Deribit's nearest-expiry options are the natural next benchmark; a spot check shows their implied volatility well below DVOL (22.7% vs 36.3% on 6 October 2026).
- **Access:** Polymarket has been blocked in Spain since 27 May 2026 under a precautionary measure by the Ministry of Consumer Affairs. This is a passive analysis of public data; no positions were taken.

5. Correction log

6 October 2026. The first version sampled Polymarket prices every 10 minutes, so each snapshot compared prices about 10 minutes old with a current BTC price. That penalized Polymarket and produced a spurious loss in the final hour. With minute-level prices the loss disappears and the 24 h and 12 h results become stronger.

6. Conclusion

A day ahead, the prices of Polymarket's daily markets contain a more accurate forecast of Bitcoin volatility than the options market's benchmark index, even after adjusting it for time of day. The edge fades as the close approaches and disappears in the final hour, when all relevant information is already in the price. This is evidence that prediction markets aggregate useful information about risk, not just direction.

The code, the full method and instructions to reproduce every result are in the project repository.